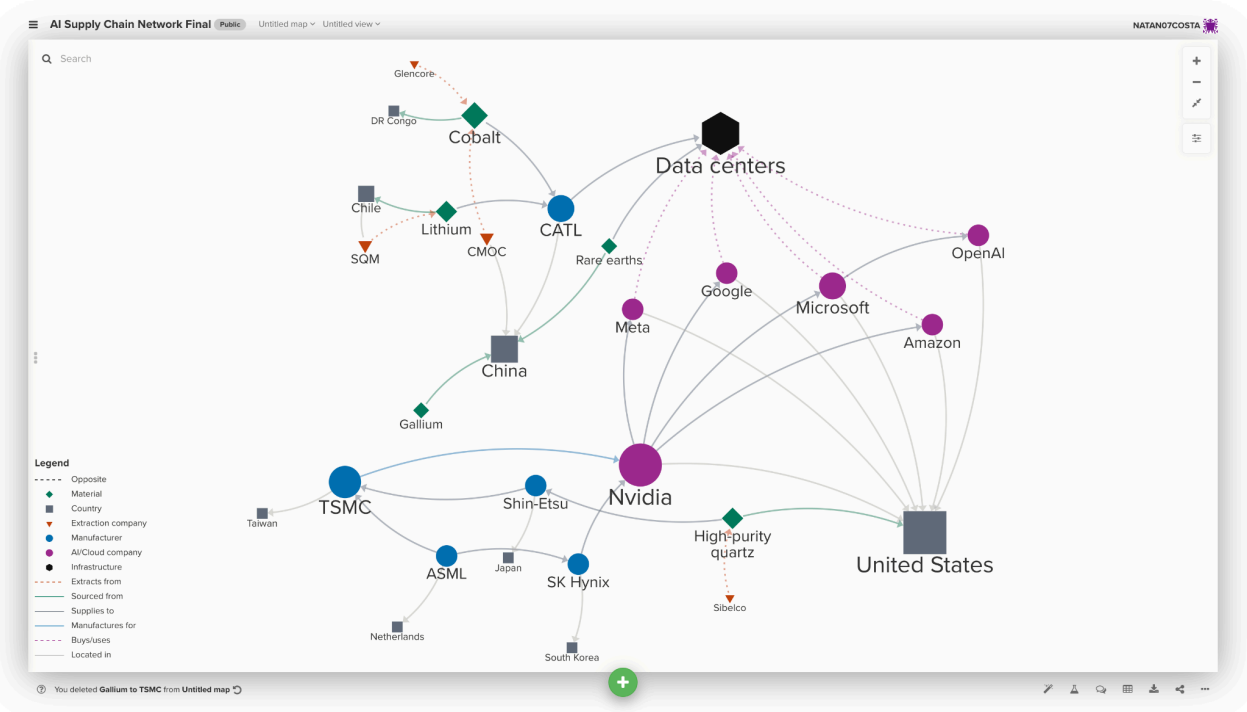
Make 7: Networks of Knowledge & Power



https://kumu.io/natan07costa/ai-supply-chain-network-final#untitled-map

Network Visualization

Tool: Kumu. The map has 29 nodes and 42 edges. Nodes are colored and shaped by type (material, country, extraction company, manufacturer, AI/cloud company, infrastructure). Edges use six relationship types, distinguished by color and line style: extracts from, sourced from, supplies to, manufactures for, buys/uses, and located in. Node size shows the number of connections.

Network Analysis

The most central node is Data centers. It ties for the highest degree centrality (7 connections, along with Nvidia and the United States) and has the highest betweenness (0.51) and closeness (0.38) in the network. Minerals, batteries, chips, and cloud companies all lead to it. Nvidia comes next in importance: it also has 7 connections and a betweenness of 0.35, and it links the chip makers (TSMC, SK Hynix) to every major cloud company. TSMC has the highest betweenness among chip makers (0.20).

Three clusters emerged: a mineral cluster (cobalt, lithium, Glencore, CMOC, SQM, DR Congo, Chile), a chip manufacturing cluster (ASML, TSMC, Shin-Etsu, SK Hynix, quartz), and a Big Tech and cloud cluster (Microsoft, Google, Amazon, Meta, OpenAI). The United States acts as a hub for the third cluster.

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What surprised me is how thin the periphery is. DR Congo, Taiwan, the Netherlands, South Korea, Japan, and the mining companies have only one or two connections, so they look small on the map, yet the real chain depends on them: without cobalt from DR Congo or chips made in Taiwan, nothing above them works. I was also surprised that CATL, a battery company, has the second-highest betweenness (0.43), because it is the main bridge between the mineral cluster and the data centers. That result partly reflects my own choices about what to include, which shows that a network is shaped by whoever builds it.

What is missing: the workers who mine cobalt, including child labor concerns in the DRC; local communities; governments and export controls; the electricity and water that data centers consume; and companies such as Samsung, Intel, Apple, AMD, and Anthropic. The network also treats every line as equal, even though a small supplier and a giant customer are not equal partners.

Artist Statement

My network shows that power in AI is concentrated in a few chokepoints. ASML is the only company that makes the EUV machines needed for the most advanced chips, TSMC fabricates the most advanced AI chips, and Nvidia designs the GPUs that nearly every cloud company buys. A few companies sit in the middle of the picture while many others hang from the edges. I chose this structure because supply chains are usually told from the top, starting with the brands people recognize.

Who is visible? Nvidia, Microsoft, Google, and OpenAI: names in the news and on stock tickers. Who is invisible? The miners in DR Congo and Chile, the countries that appear only as small dots, and the workers, land, water, and electricity behind every line. In my visualization, DR Congo looks as small as Japan, even though cobalt mining carries enormous human and environmental costs. A visualization can hide power by making all nodes and lines look equal.

A different structure would tell a different story. If I mapped labor instead of companies, mine workers and data center workers would be at the center and Nvidia would be at the edge. If I mapped environmental impact, data centers and mining regions would become the hubs. If I sized nodes by profit, the mining nodes would shrink even more. Every network is a choice about who gets to be a node, and what I leave out is also a statement about power.